

MARK SCHEME for the October/November 2007 question paper

9709 MATHEMATICS

9709/05

Paper 5, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2007 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only - often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR -1	A penalty of MR -1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through $\sqrt{}$ " marks. MR is not applied when the candidate misreads his own figures - this is regarded as an error in accuracy. An MR-2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA -1	This is deducted from A or B marks in the case of premature approximation. The PA -1 penalty is usually discussed at the meeting.

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1	(i)	$T = 4 \times 0.25 / 0.25$ or $4 \times 0.5 / 0.5$	M1		For using $T = \lambda x / L$	
		Tension is 4N	A1	2		
	(ii)		M1		For using Newton's second law	
		$2 \times 4 \times 0.6 = 0.6a$	A1ft			
		Acceleration is 8ms^{-2}	A1	3		5

2	(i)		M1		For using $a = v^2/r$ and Newton's second law horizontally	
		$T \sin 30^\circ = 0.4 \times 0.6^2 / 0.08$	A1			
		Tension is 3.6N	A1	3		
	(ii)		M1		For resolving forces vertically (3 terms)	
		$R + T \cos 30^\circ = 0.4g$	A1			
		Force is 0.882N	A1ft	3	ft $[4 - \text{candidate's } T \cos 30^\circ]$ (must be +ve) or $T = 2.96$ from consistent sin/cos mix	6

3	(i)		M1		For taking moments about A (3 terms)	
		$100x(1 \cos \alpha) + 300x(2 \cos \alpha) = T \times 0.7$	A1		α is the angle made by the string with the vertical	
		where $\cos \alpha = 0.96$	A1			
		Tension is 960N	A1ft	4	ft $1000 \cos \alpha$	
	(ii)	$X = 268.8$ (269)	B1ft		ft $1000 \sin \alpha \cos \alpha$	
		$Y + 10g + 300 = 960 \cos \alpha$	M1		For resolving forces vertically (4 terms)	
		$Y = 521.6$ (522)	A1	3		7

4	(i)	$0.4g - 0.08v = 0.4a$	M1		For using Newton's second law	
		Acceleration is $10 - 0.2v$	A1	2		
	(ii)	$\int \frac{dv}{50 - v} = \int 0.2 dt$	M1		For using $a = dv/dt$, separating the variables and attempting to integrate	
		$-\ln(50 - v) = 0.2t (+ C)$	A1			
		$-\ln(50 - v) = 0.2t - \ln 50$	M1		For using $v(0) = 0$ to find C	
		$50 - v = 50e^{-3}$	M1		For substituting $t = 15$ and solving for v	
		Speed is 47.5ms^{-1}	A1	5		7

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5	(i)		M1		For using $EE = \lambda x^2/2L$	
			M1		For using $EE_{S2}(\text{initial}) = \frac{1}{2}mv^2$ + EE_{S1} (S2 just slack)	
		$\frac{1}{2}(16 \times 1^2/0.5) =$ $\frac{1}{2}0.5v^2 + \frac{1}{2}(16 \times 0.6^2/0.4)$	A1			
		Speed is 5.93ms^{-1}	A1	4		
	(ii)		M1		For using $EE_{S2}(\text{initial}) = EE_{S1}$	
		$\frac{1}{2}(16 \times 1^2/0.5) = \frac{1}{2}(16x^2/0.4)$ ($x = 0.894$)	A1			
		Distance is 1.29m	A1	3		7

6	(i)	$3 = 8 \tan 35^\circ - g8^2/(2V^2 \cos^2 35^\circ)$	M1		For substituting $\theta = 35^\circ$, $x=8$ and $y=3$ into the trajectory formula or eliminating T from $8 = Vt \cos 35^\circ$, $3 = Vt \sin 35^\circ - \frac{1}{2}gT^2$	
			M1		For solving for V	
		Speed is 13.5ms^{-1}	A1			
OR		For eliminating VT from $8 = Vt \cos 35^\circ$, $3 = Vt \sin 35^\circ - \frac{1}{2}gT^2$ to find T ($=0.721$)	(M1)			
		For back substituting to find V	(M1)			
		Speed is 13.5ms^{-1}	(A1)			
			M1		For substituting $\theta = 35^\circ$, $x=8$ and value of V into $x = Vt \cos \theta$ or stating value of T found in (i) (alternate method)	
		T = 0.721	A1	5		
	(ii)		M1		For using $v_x = V \cos 35^\circ$ and $v_y = V \sin 35^\circ - gT$	
			M1		For using $\tan \alpha = v_y/v_x$	
		$\tan \alpha = 0.55(22) \div 11(.09)$	A1		May be implied by final answer	
		Direction 2.85° to the horizontal	A1	4	Accept 2.8 or 2.9	
OR			(M1)		For differentiating the trajectory equation w.r.t.x	
		$y' = \tan 35^\circ - gx/(V^2 \cos^2 35^\circ)$	(A1)			
			(M1)		For using $\tan \alpha = y'(8)$ (0.0498)	
		Direction 2.85° to the horizontal	(A1)			9

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7	(i)		M1		For taking moments	
		$(20 \times 30) \times 10 - (15 \times 20) \times 12.5 = (20 \times 30 - 15 \times 20)y$ or $2x(20 \times 5) \times 10 + (5 \times 20) \times 2.5 = [2x(20 \times 5) + (5 \times 20)]y$	A1			
		$y = 7.5$	A1	3		
	(ii)	$\tan \alpha = y/(DE/2)$	M1		On the point of toppling when C is vertically above E used	
		$\tan \alpha = \frac{1}{2}$	A1			
		For using $\mu > F/R = \tan \alpha$ to obtain printed result	B1	3	$F/R = \tan \alpha$ may be quoted or found using $F = W \sin \alpha$, $R = W \cos \alpha$	
	(iii)	$\tan \beta = (20 - y)/15$	B1		β is the angle that toppling would take place	
			M1		For using $\mu = \tan \theta$ (may be quoted) and $\theta < \beta$, where θ is the angle at which the prism slides	
		$\mu < \frac{5}{6}$ (AG)	A1	3		9